

Computer Netzwerke und verteilte Systeme Summer 2026



Exercise 7: Flow Control

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Deadline: The deadline for this exercise is **09.06.2026, 23:55** and in total **32 bonus points** can be achieved.

Bonus points: *Computer Netzwerke und verteilte Systeme* features a bonus point system to acknowledge first-rate performance. Exercise points count as bonus. Throughout the entire course, a bonus of up to a maximum of 0.7 can be obtained. The bonus does not help in passing the exam: You need to pass the exam with a 4.0 or better before any bonus is applied.

Group size: The recommended group size for submissions is 3. You may also submit your exercises as a group of 2 or alone if you do not find a group.

Important: Every theoretical task and subtask require proper sources. Submissions without proper sources will automatically receive 0 points.

For most of the task the slides of the lecture will be enough. For **research** task, use the materials provided in the lecture, or the internet.

Refrain from copying and pasting your answers, as this is considered plagiarism and will have **consequences**, as will using AI tools to create your answers. Instead, explain your answers in your own words and cite your sources!

Write down your answers in a legibly and unambiguously way, either by creating formatted documents (LaTeX, Word, typst, etc.) or by handwriting (**ensure that your handwriting is legible**).

Refrain from using colors that create less contrast between the text and the background. For marking tasks you may use blue and orange.

Task 1: Group Exercise: Flow Control

- (a) Briefly explain the main task of Flow Control.
- (b) Which problems may occur when providing Flow Control in the Transport Layer? **State two.**
- (c) What is the Difference between Rate-based and Window-based Flow Control? Which of these would you use in a High-Speed Network?

Task 2: Group Exercise: TCP/Flow Control

A sender transmits **6 frames** to a receiver (sequence numbers **0 – 5**) using the sliding window method. The parties have agreed on a **window size of 2**. Frames are acknowledged by adding the corresponding sequence number to the ACK messages (each frame is acknowledged individually). The time to transmit one frame (as well as one ACK) is defined as 1. **Sending and receiving frames generally has higher priority than producing or receiving ACKs.** One station cannot send and receive at the same time. Fill in every cell Table 1, assuming all frames are transmitted correctly and without errors.

t	Sender window	Frame(s) ready	Sender sending	Sender receiving	Receiver receiving	Receiver sending	Receiver window
0	0-0	-	-	-	-	-	0-2
1	0-1	F0	F0	-	-	-	0-2
2	0-1	-	-	-	F0	-	1-2
3	0-2	F1	F1	-	-	ACK0	1-3
4	1-2	F2			F1		
5		F2, F3					
6		F3, F4					
7		F3, F4					
8		F4					
9		F4					
10		-					
11		-					
12		F5					
13		-					
14		-					
15		-					
16		-					

Table 1: Template table for task 2

Task 3: Group Exercise: Sliding Window Protocols

In sliding window protocols, the sender maintains a set of sequence numbers corresponding to frames that have not been acknowledged yet. The receiver also maintains a receiving window corresponding to the frames it is permitted to accept. The sequence numbers within the sender's window represent frames that have been sent but are not yet acknowledged. The sequence numbers within the receiver's window are increased as soon as an acknowledgement is sent.

Assume there is a sliding window protocol as shown. A sender transmits **6 frames** to a receiver (sequence numbers **0 – 5**) using the sliding window protocol. The parties have agreed on a **window size of 4**. Sender sends one frame at a time, if the current window size allows. Each frame is acknowledged individually. **Sending and receiving frames generally has higher priority than producing or receiving ACKs**. One station cannot send and receive at the same time. There is a **one-step transmission delay** from the sender to the receiver. Draw the sliding windows as shown below (no transmission errors) and state which packets are transmitted in each step (e.g., “send 0”, “send ack 2”, “rec ack 2”, “wait”). Given is the initial state. Assume that all the frames are ready in the initial state.

0	1	2	3	4	5	6	7
init							

Table 2: Sender table part 1 for task 3

0	1	2	3	4	5	6	7
init							

Table 3: Receiver table part 1 for task 3

8	9	10	11	12	13	14

Table 4: Sender table part 2 for task 3

8	9	10	11	12	13	14

Table 5: Receiver table part 2 for task 3

Task 4: Congestion Control (4 Points)

- (a) Explain the fundamental limitation of open-loop congestion control compared to closed-loop. Illustrate your explanation with a concrete network scenario. (1.5 P)
- (b) Why do routers employ a **warning state** (mentioned in the lecture) instead of using only one simple maximum threshold that is defined as the maximum network capacity? Name (no explanation needed) two actions that can be taken when a router enters the warning state. (2.5 P)

Attention: Submissions without references will not be graded! Please add sources.

Task 5: TCP/Flow Control (14 Points)

A sender transmits **9 frames** to a receiver (sequence numbers **0 – 8**) using the sliding window method. The parties have agreed on a **window size of 5**. Frames are acknowledged by adding the corresponding sequence number to the ACK messages (each frame is acknowledged individually). The time to transmit one frame (as well as one ACK) is defined as 1. **Sending and receiving frames generally has higher priority than producing or receiving ACKs**. One station cannot send and receive at the same time. Fill in every cell in Table 6, assuming all frames are transmitted correctly and without errors.

t	Sender window	Frame(s) ready	Sender sending	Sender receiving	Receiver receiving	Receiver sending	Receiver window
0	0-0	-	-	-	-	-	0-5
1	0-1	F0	F0	-	-	-	0-5
2	0-2	F1	F1	-	F0	-	1-5
3	0-2	-	-	-	F1	-	2-5
4	0-3	F2					
5		F3					
6		F4					
7		F5					
8		F5					
9		-					
10		-					
11		F6					
12		-					
13		-					
14		F7					
15		-					
16		-					
17		-					
18		F8					
19		-					
20		-					
21		-					

Table 6: Template table for task 5

Task 6: Sliding Window Protocols (14 Points)

In sliding window protocols, the sender maintains a set of sequence numbers corresponding to frames that have not been acknowledged yet. The receiver also maintains a receiving window corresponding to the frames it is permitted to accept. The sequence numbers within the sender's window represent frames that have been sent but are not yet acknowledged. The sequence numbers within the receiver's window are increased as soon as an acknowledgement is sent.

Assume there is a sliding window protocol as shown. A sender transmits **6 frames** to a receiver (sequence numbers **0 – 5**) using the sliding window protocol. The parties have agreed on a **window size of 3**. Sender sends one frame at a time, if the current window size allows. Each frame is acknowledged individually. **Sending and receiving frames generally has higher priority than producing or receiving ACKs.** One station cannot send and receive at the same time. There is a **one-step transmission delay** from the sender to the receiver. Draw the sliding windows as shown below (no transmission errors) and state which packets are transmitted in each step (e.g., “send 0”, “send ack 2”, “rec ack 2”, “wait”). Given is the initial state. Assume that all the frames are ready in the initial state.

0	1	2	3	4	5	6	7
init							

Table 7: Sender table part 1 for task 6

0	1	2	3	4	5	6	7
init							

Table 8: Receiver table part 1 for task 6

8	9	10	11	12	13	14

Table 9: Sender table part 2 for task 6

8	9	10	11	12	13	14

Table 10: Receiver table part 2 for task 6